THE KERNEL OF THE MAP TO A FINITE LEVEL IS THE F-ADIC POWER OF THE LIMIT, SO THE LIMIT TOPOLOGY IS F-ADIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one cross-chapter label, recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- The f-divisibility condition on the inverse system. Stacks Project, Cohomology of Sheaves, Lemma lemma-equivalent-f-good (tag 0H39), condition (1). Gives the identification F_n = F_{n+1}/f^n F_{n+1} that turns the connecting maps into multiplication by f^c.

Source reference. Original lemma 0H39: sources/stacks-chow/tags.txt, source lines 21309–21309. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Suppose X, f, $(\mathcal{F}_n)$ satisfy condition (1) of Lemma 0H39. Let $p \geq 0$ and set $H^p = \lim H^p(X, \mathcal{F}_n)$. Then $f^c H^p$ is the kernel of $H^p \to H^p(X, \mathcal{F}_c)$ for all $c \geq 1$. Thus the limit topology on H^p is the f-adic topology.*

Proof. Let $c \geq 1$. It is clear that $f^c H^p$ maps to zero in $H^p(X, \mathcal{F}_c)$. If $\xi = (\xi_n) \in H^p$ is small in the limit topology, then $\xi_c = 0$, and hence ξ_n maps to zero in $H^p(X, \mathcal{F}_c)$ for $n \geq c$. Consider the inverse system of short exact sequences

$$0 \to \mathcal{F}_{n-c} \xrightarrow{f^c} \mathcal{F}_n \to \mathcal{F}_c \to 0$$

and the corresponding inverse system of long exact cohomology sequences

$$H^{p-1}(X, \mathcal{F}_c) \to H^p(X, \mathcal{F}_{n-c}) \to H^p(X, \mathcal{F}_n) \to H^p(X, \mathcal{F}_c)$$

Since the term $H^{p-1}(X, \mathcal{F}_c)$ is independent of n we can choose a compatible sequence of elements $\xi'_n \in H^p(X, \mathcal{F}_{n-c})$ lifting ξ_n. Setting $\xi' = (\xi'_n)$ we see that $\xi = f^c \xi'$ as desired. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.